

TLAMATINI

Complete Project Dossier: what the system does, how it works, how to use Tlamatini, complete tracked file tree, and effective line inventory



Measure	Value
Generated	May 26, 2026 11:49 PM UTC-0600
Current HEAD	b9ad650 - Implementing STMer agent, now Tlamatini is able to program completely in hardware of STM32F proc
Resolved version	1.9.0 (git)
Tracked files	505
Workflow agents	68
Multi-Turn tools	75
Skills	24
Total effective lines	111,898
Total physical text lines	152,810

1. What Tlamatini Is

- Tlamatini is a local-first AI developer assistant built with Django, Django Channels, LangChain, LangGraph, FAISS/BM25 retrieval, and a large in-repository agent application.
- She combines a browser chat surface, a Retrieval-Augmented Generation stack, a Multi-Turn tool executor, MCP-backed context providers, wrapped chat-agent runtimes, and a visual Agentic Control Panel for workflow design.
- She is designed for development operations: codebase analysis, file and directory context, command execution, Python execution, screenshots, web/search helpers, notifications, DevOps tools, local model operation, Windows packaging, first-person self-knowledge about her own runtime, and now critical-mission STM32F4 firmware control.

What the system does

- Answers codebase questions with loaded file or directory context.
- Uses hybrid retrieval to extract metadata, split content, rank source chunks, and respect context budgets.
- Gives operators GUI-first database maintenance through the new DB dropdown for backup and staged database replacement.
- Warns GPU-host operators before a directory-context load is likely to saturate VRAM and degrade embedding throughput.
- Exposes a coherent versioning surface across builds, runtime UI, logs, and an open health-check endpoint.
- Carries a first-person self-knowledge map so she can answer more accurately about her own architecture, ports, runtime modes, pages, and capabilities.
- Can command Kali Linux offensive-security tooling through MCP-Kali-Server for authorized recon, enumeration, web scanning, and assessment workflows.
- Can scaffold, author, build, flash, reset, and observe STM32F4 firmware through STM32er and the STM32 Template Project MCP, with a fail-safe preflight before any hardware mutation.
- Can manage real desktop windows by title: focus them, tile them, resize them, list them, and close them deterministically through Win32 calls.
- Can drive a real Playwright browser through scripted interactive steps for logins, forms, assertions, downloads, extraction, and end-to-end UI checks.
- Can drive a live Unreal Engine 5 editor through the Unreal MCP plugin, from either Multi-Turn chat or the visual workflow canvas.
- Actively reaps orphaned Windows console-host and pool-child processes so long Multi-Turn or ACPX sessions do not leave misleading Tlamatini-icon ghosts in Task Manager.
- Runs checked Multi-Turn requests through request-scoped planning, capability selection, tool calls, observations, monitoring, and final synthesis.
- Can optionally inspect and modify her own bundled source tree in self-modify builds after verifying that ``TlamatiniSourceCode/`` is actually present.
- Launches wrapped copies of selected workflow agents in isolated runtime folders without mutating templates.
- Lets users design, validate, save, pause, resume, and stop visual workflows through the Agentic Control Panel.
- Turns successful Multi-Turn tool executions into starter ``.flw`` workflows that can be inspected and validated in ACP.
- Packages the project into a distributable Windows release with installer and uninstaller tooling.

How it works

- Browser UI sends chat and workflow requests through Django views and Channels WebSockets.
- RAG chains load selected file/directory context, retrieve relevant chunks, and build answer prompts.
- DB-menu actions validate directories or SQLite files in the browser, then call Django views that either copy the live database out or stage a replacement into ``DB/ToLoad/db.sqlite3``.
- Before a heavy directory embedding run on supported NVIDIA hosts, a fail-open pre-flight guard can estimate VRAM pressure and surface a non-blocking warning in chat.

- Version resolution now flows through git tags, a runtime resolver module, generated build artefacts, and an open ``/agent/version/`` endpoint.
- A first-person self-knowledge file (``Tlmatini.md``) is injected into prompt construction for all chains, but loaded user context still outranks that self-reference when the request is a generic summary of the provided project.
- When Multi-Turn is enabled, the global planner selects context and tool stages before the executor binds only the relevant tools, including wrapped deterministic agents such as De-Compressor.
- Optional self-modify builds bundle ``TlmatiniSourceCode/``; prompt rules require her to verify that directory exists before claiming she can inspect or change her own code.
- The Kalier path talks directly to the MCP-Kali-Server Flask API over HTTP with Python-stdlib ``urllib``, auto-seeding the default box from ``kali_server_url`` in Config -> URLs before any one-off per-call override is applied, and captures one atomic ``INI_SECTION_KALIER`` block per run.
- The STM32er path spawns the STM32 Template Project MCP stdio server, performs the MCP initialize handshake, runs exactly one requested tool or composite action, and emits one atomic ``INI_SECTION_STM32ER`` block with the result, project directory, and stage metadata.
- Before any flash-capable STM32er action, a critical-mission preflight validates the arm-none-eabi toolchain, STM32CubeIDE, programmer path, ST-LINK presence, and STM32F-family match; compile-only steps can run boardless, but unsafe hardware mutations are refused fail-safe.
- The Windower path uses Win32 APIs plus the cross-process ``AttachThreadInput`` focus-transfer dance to locate windows by title and apply one lifecycle action while still returning structured geometry/state fields.
- The Playwrighter path loads a declarative step list, drives Playwright against Chromium/Firefox/WebKit, and emits one atomic ``INI_SECTION_PLAYWRIGHTER`` block with status, assertions, extracted values, and the final URL.
- The Unreal path opens a TCP socket to the Unreal MCP plugin, sends one ``{"type": "command", "params": {...}`` payload, captures the JSON reply, and emits one ``INI_SECTION_UNREALER`` block for downstream logic.
- After spawn-capable tool calls and again after the final answer, the orphan reaper can sweep dead descendants, orphaned ``conhost.exe`` companions, and stale pool-linked processes without ever raising into the chat path.
- Tool calls execute in the backend, append observations, and may create wrapped runtime copies under ``agent/agents/pools/_chat_runs/``.
- On the next full start-up, ``manage.py`` can swap a staged database into place before Django imports, while archiving the previous live database under ``DB/Older/<timestamp>``.
- ACP flows deploy session-scoped pool instances, wire config values, validate NxN graph rules, and execute through Starter-driven flow semantics.
- Build scripts collect static assets, bundle Django/Python resources, add agent templates, and assemble ``pkg.zip``, ``Uninstaller.exe``, and ``dist/Tlmatini_Release/``.

2. Architecture Layers

Layer	Role
Browser interfaces	Chat page plus Agentic Control Panel templates and JavaScript modules.
Django/Channels	Authentication, views, WebSockets, session state, message persistence, and ASGI startup.
RAG and context	Metadata extraction, text splitting, FAISS/BM25 retrieval, context budgeting, and fallback behavior.
Multi-Turn engine	Capability registry, global execution planner, explicit tool loop, answer parsing, and answer-success classification.
Tools and agents	Core tools, MCP context providers, wrapped chat-agent launchers, and the current visual workflow agent templates.
Packaging	PyInstaller build scripts, shortcut registration, `.flw` association, installer, uninstaller, and release folder assembly.

Design principles

- Evidence-first answers: Tlamatini grounds responses in selected project context and hybrid retrieval rather than freeform model memory.
- Explicit orchestration: checked Multi-Turn uses a visible tool loop, capability scoring, and staged planning instead of a single opaque call.
- Operational reversibility: risky changes such as database replacement are staged, archived, and delayed to the only safe window instead of hot-swapped mid-session.
- Fail-open diagnostics: GPU pressure probes and session-restore safeguards warn early without breaking CPU-only or degraded environments.
- Runtime isolation: wrapped chat-agent copies run in session-scoped folders so template agents remain pristine while live runs stay inspectable.
- Self-knowledge with scope discipline: she can talk accurately about herself without letting self-reference override a user-loaded project context.
- Operator truth over vibes: Exec Report tables, tlamatini.log, skill audits, and ACPX transcripts make the system auditable after execution.

3. Installation, Configuration, and Everyday Use

Installation essentials

- Python 3.12.10 is the strongly recommended source-mode version in the README, and the codebase has been tested most deeply there.
- Source installs require a clone, virtual environment, dependency install from `requirements.txt`, migrations, a superuser, static collection, and then the web server.
- You can run either the checked-in cloud/back-end defaults from `Tlmatini/agent/config.json` or a local Ollama-backed configuration with matching model names.
- Packaged Windows installs create a default `user / changeme` account; manual source installs use your own `createsuperuser` account instead.

Configuration essentials

- Source mode resolves `Tlmatini/agent/config.json`; frozen builds resolve `config.json` next to the executable; `CONFIG\_PATH` overrides both.
- Core keys include `embedding-model`, `chained-model`, `ollama\_base\_url`, `ollama\_token`, `enable\_unified\_agent`, `unified\_agent\_model`, and `unified\_agent\_max\_iterations`.
- URL configuration now also includes `kali\_server\_url` plus the STM32er bootstrap fields `stm32\_mcp\_server\_script`, `stm32\_mcp\_python`, `stm32\_template\_dir`, `stm32\_ide\_root`, `stm32\_mcp\_repo\_url`, and `stm32\_mcp\_install\_dir`, all edited from `Config -> URLs` and inherited automatically by the chat-side wrapped tools.
- The chat-side Config -> Models and Config -> URLs dialogs are now first-class configuration surfaces, and they can explicitly ask the operator to reconnect when saved values change live-session assumptions.
- The separate DB dropdown is not a config editor: it is a maintenance surface for copying the live SQLite database out or staging a replacement for the next full start-up.
- Multi-Turn is toggled from the chat toolbar, but it depends on the unified-agent configuration and the selected model/base-url pairing being valid; the current default iteration ceiling is 4096.
- Image interpretation can run through Claude-backed cloud paths or Qwen/Ollama-backed local paths, and remote Ollama can be protected with a bearer token.

DB menu and database swap-in

- The new DB dropdown gives operators two GUI-first maintenance paths: `Backup database` copies the live SQLite file out, and `Set DB` stages a chosen `db.sqlite3` for the next full start-up.
- Both dialogs are live-validated in the browser and now expose Browse buttons that open native host-side pickers for folders or `db.sqlite3` files.
- Backup is read-only and uses the currently live database path; Set DB never hot-swaps the file mid-session because Django already holds SQLite open.
- Set DB writes the selected file to `DB/ToLoad/db.sqlite3`; the real swap happens only at the top of `manage.py` before Django imports anything.
- When that swap runs, the previous live database is moved into `DB/Older/<timestamp>/db.sqlite3`, creating a built-in rollback trail instead of overwriting history.
- Reconnect is not enough for this path: the operator must fully restart Tlmatini so the pre-Django swap window opens again.
- If the staged file is bad or locked, startup fails open: Tlmatini logs the error and continues with the previous live database.

Versioning system

- Tlmatini now follows Semantic Versioning 2.0.0 with git tags as the single source of truth: you tag, then you build, instead of hand-editing version strings across files.

- The build path resolves a version once and propagates it into generated runtime metadata, Win32 VERSIONINFO resources, and the release-folder naming convention.
- On untagged commits the resolver falls back honestly to a git-derived development version instead of failing the build or lying about the release state.

Version surfaces

- Operators can now see the running version in the About dialog, the startup banner, the open ``GET /agent/version/`` health-check endpoint, and Windows file properties on the built executables.
- The runtime resolver lives in ``Tlamatini/agent/version.py``, while the build-oriented coordination logic lives in the repo-root ``versioning.py`` and the longer policy notes live in ``VERSIONING.md``.
- Build overrides are supported in a predictable order: CLI ``--version``, then ``TLAMATINI_VERSION``, then git describe, then the sentinel ``0.0.0+unknown``.

Self-knowledge in v1.8.0

- Version ``1.8.0`` gives Tlamatini a first-person self-knowledge map in ``Tlamatini/agent/Tlamatini.md``, so she can answer more accurately about her own architecture, runtime modes, open ports, pages, and internal capability surface.
- That self-reference is injected into all prompt chains through ``prompt.pmt`` and ``agent/rag/config.py``, but it fails open if the file is missing or unreadable and it never overrides a user-loaded project when the request is a generic summary of the provided context.
- The language contract matters too: when a pronoun is used for Tlamatini in the documentation or prompt guidance, she is referred to as ``she`` / ``her``, matching the updated handbook identity rules.

Self-modify builds

- Self-modification is a separate capability axis from frozen-vs-source runtime: only builds created with ``python build.py --self-modify`` bundle ``TlamatiniSourceCode/`` next to the application.
- When that directory is present, she can inspect and modify her own shipped source tree; when it is absent, she must say so plainly and fall back to her injected self-knowledge plus the surrounding docs.
- The build pipeline now announces that choice explicitly, so operators can tell whether a release is self-modify-capable before asking her to work on herself.

Multi-Turn 4096-turn autonomy

- The unified-agent loop now defaults to 4096 iterations instead of 256, giving long autonomous operator runs much more room before they exhaust the turn budget.
- That expansion is about conversational/tool-loop depth, not about blindly firing more tools at once: the planner's selected-tool cap still keeps the tool surface bounded per request.
- Operationally, the bigger ceiling helps long workflows, while duplicate-call guards and the dedicated ``chat_agent_sleeper`` tool remain the antidote to accidental busy-polling loops.

De-Compressor agent

- De-Compressor is the deterministic archive worker for compression and decompression tasks, deciding direction from whichever side exposes a recognized archive extension.
- Supported archive families are ``.gz``, ``.zip``, ``.7z``, ``.tar.gz``, and ``.gz.tar``; file-to-folder extraction and file-or-directory packing are both documented in the README and Book.
- Password handling is explicit: ``passwordless=true`` skips it, while ``passwordless=false`` requires the ``DE_COMPRESSER_PWD`` environment variable and fails fast if the secret is missing.

De-Compressor integration and fallback behavior

- The agent is reachable from both operator surfaces: ACP canvas nodes wire through dedicated connection-update views, and checked Multi-Turn can invoke it through ``chat_agent_de_compressor``.

- Format engines are practical rather than magical: `stdlib`gzip`` and `zipfile`` cover core cases, ``7z`` is preferred for encrypted ``7z`` or ``zip``, and ``py7zr`` was added to ``requirements.txt`` as the Python fallback.
- Every run emits an ``INI_SECTION_DE_COMPRESSER<<< ... >>>END_SECTION_DE_COMPRESSER`` block and still triggers ``target_agents``, so downstream Parametrizer or Raiser logic can branch on ``success=true|false`` instead of guessing from prose.

Unreal MCP and the Unrealer agent

- Unreal MCP is a UE5 plugin that runs inside the editor and listens on ``127.0.0.1:55557`` for one JSON command per TCP connection; Tlamatini is the client side of that link, not the plugin host.
- The Unrealer workflow agent forwards any command the connected plugin build exposes, up to a 53-command surface across nine categories: actor manipulation (incl. viewport screenshots), Blueprint authoring and node wiring, input mappings, UMG widget building, in-editor Python/console execution, level I/O, asset import, and material authoring. Headless build/cook/test is out of scope (it needs `UnrealEditor-Cmd``, not the editor socket).
- The same integration is available in both operator surfaces: checked Multi-Turn via ``chat_agent_unrealer``, and ACP canvas flows via the visual Unrealer node plus Parametrizer chaining.

Extended Unreal MCP surface

- Unrealer's documentation now points at the public ``XAIHT/XaihtUnrealEngineMCP`` fork, the Unreal Engine MCP variant developed specifically for Tlamatini.
- That fork exposes the extended 53-command, nine-category surface documented in README and Book: not only the base editor/Blueprint/UMG verbs, but also system, level, asset, material, screenshot, viewport, and in-editor Python paths.
- The practical message for operators is simple: Unrealer forwards the connected plugin's verb surface directly, so Tlamatini's client does not need a new release every time the plugin adds another supported command.

Installing the UE5 plugin and smoke-testing it

- Install the upstream plugin into `<YourProject>/Plugins/UnrealMCP/``, enable it from ``Edit -> Plugins``, restart UE5, and confirm the Output Log says ``UnrealMCP listening on 127.0.0.1:55557``.
- Tlamatini does not compile or embed the plugin; the Unreal editor must already be running with the listener bound before any Unrealer call can succeed.
- A seeded smoke test ships in the Prompts table: ``Unreal MCP End-to-End Editor Drive`` walks through sanity-check, actor spawn, Blueprint creation/compile, and UMG widget assembly.

Orphan-process cleanup

- Tlamatini now ships a three-tier orphan reaper in ``Tlamatini/agent/orphan_reaper.py`` focused on Windows console-host leftovers such as ``conhost.exe`` and ``openconsole.exe``.
- Tier 1 runs after spawn-capable Multi-Turn tool calls, Tier 2 runs once after the final answer in a background thread, and Tier 3 runs again during application shutdown through the same cleanup path that already tears down pools.
- The candidate set stays narrow on purpose: dead descendants of the current process tree, orphaned console hosts tied to that tree, and pool-linked processes whose ``cmdline`` still points into ``agents/pools/...`` even though tracking records are gone.

Orphan-process prevention and survivor reporting

- Prevention landed alongside cleanup: Windows spawn sites now use ``CREATE_NO_WINDOW | DETACHED_PROCESS | CREATE_NEW_PROCESS_GROUP`` with stdio piped to ``DEVNULL``, so the console host is usually never allocated in the first place.
- The ACPX runtime now kills full process trees instead of only top-level wrappers, and pool-agent scripts gained a conservative ``subprocess.Popen`` guard so forgotten descendants still inherit ``CREATE_NO_WINDOW``.
- If something survives both cleanup tiers, Tlamatini sends a second chat bubble listing each surviving ``name + PID``; the reaper never raises into the user path because a cleanup crash would be worse than the leftovers it tried to remove.

How to use it

- Run from source: create a virtual environment, install requirements, migrate, create a superuser, collect static files, and start Django.
- Open `/agent/` for chat. Load a file or directory context before asking codebase-specific questions.
- Keep Multi-Turn unchecked for direct Q&A; enable Multi-Turn for tasks that need tools, wrapped agents, monitoring, or workflow seeding.
- If you want her to inspect or modify herself, verify that `TlamatiniSourceCode/` exists in the current build first; self-modify is optional and absent builds must be treated honestly as read-only about their own code tree.
- For authorized Kali Linux assessments, run MCP-Kali-Server on the Kali box, set `Config -> URLs -> Kali server (Kali)` once, and then call `chat_agent_kalier` from Multi-Turn with the desired `action` and `target` without repeating the box URL each turn.
- For STM32 firmware work, install STM32CubeIDE, leave `Config -> URLs -> STM32 MCP server script` blank for zero-config bootstrap, and then call `chat_agent_stm32er` from Multi-Turn with one `action` at a time such as `validate`, `create_project`, `write_source`, `build`, `build_and_flash`, `serial_session`, or `live_monitor`.
- For desktop-window control, call `chat_agent_windower` from Multi-Turn to focus, tile, resize, list, or close a window by title, or model the same action in ACP with the Windower node.
- For interactive web automation, call `chat_agent_playwrighter` from Multi-Turn with a `steps_json` script, or author the same step list visually with the Playwrighter node on the canvas.
- For Unreal Engine work, enable the Unreal MCP plugin inside a live UE5 project first, then call `chat_agent_unrealer` from Multi-Turn or use the visual Unreal node on the canvas.
- Archive jobs can now be described directly in Multi-Turn or modeled visually in ACP: De-Compressor infers compress vs decompress from the `input` or `output` extension.
- If a second post-answer warning bubble ever lists surviving `name + PID` entries, treat it as an honest cleanup report and end the listed processes manually from Task Manager if needed.
- Use the `ACPX-Skills` navbar menu when you need to browse, enable/disable, diagnose, or reload the shipped SKILL.md catalog without asking the LLM to be your admin surface.
- Use the DB dropdown when you need a safe database snapshot or want to stage a different `db.sqlite3` for the next start-up without hot-swapping the live SQLite file.
- Open `/agentic_control_panel/` to drag agents, connect them, configure each node, validate, start, pause/resume, stop, and save `.flw` workflows.
- Use `python build.py`, `python build_uninstaller.py`, and `python build_installer.py` only when producing a packaged Windows release.

Agent descriptions and catalog source of truth

- The authoritative human-readable source for workflow-agent descriptions is `agents_descriptions.md` at the repo root, not an embedded JavaScript map or a hard-coded Django list.
- Current validation compares live template directories, `agents_descriptions.md` rows, and the README / Book bestiary counts so the generated dossier stays aligned with the running product.
- The ACP sidebar hover tooltip and the right-click Description dialog both resolve through that markdown file first; `README.md` is only a legacy fallback when the dedicated description file is absent.

Reviewer and Analyzer

- The workflow catalog now includes two new high-value specialists: Reviewer and Analyzer, lifting the canvas inventory to 64 agents and the seed-skill catalog to 23 SKILL.md packages.
- Reviewer is LLM-powered: it resolves a git diff for `repo_path`, reviews it with a senior-engineer prompt, and emits an `INI_SECTION_REVIEWER` block whose first routable field is `verdict = APPROVE | REQUEST_CHANGES | COMMENT`.
- Analyzer is deterministic and scanner-driven: it runs whichever of `bandit`, `semgrep`, `ruff`, `eslint`, `gitleaks`, and `pip-audit` are installed on PATH over `target_path`, then emits an `INI_SECTION_ANALYZER` block with `status` and

`total\_findings` for downstream routing.

- Both agents always trigger `target\_agents`, so a downstream Forker can branch on `{verdict}`, `{status}`, or `{total\_findings}` instead of scraping prose.
- They are intentionally canvas-only workflow agents: there is no wrapped `chat\_agent\_reviewer` or `chat\_agent\_analyzer`, and therefore no duplicate Exec Report row family for those names.
- The chat-side counterparts live in the ACPX skill catalog instead: `code-review` exposes senior-engineer git-diff review, while `security-audit` exposes the deterministic multi-scanner sweep.

Operator surface counts

- The live operator surface now stands at 68 workflow agents, 75 Multi-Turn tools, 12 ACPX tools, and 24 skills.
- Source inspection confirms the newer total: 43 wrapped chat-agent tools in `chat\_agent\_registry.py`, which combines with 20 core Python tools and 12 ACPX/Skill tools for 75 Multi-Turn tools overall.
- README.md and BookOfTlmatini.md now agree on those counts, so the dossier can mirror the docs and the live registry without reconciliation footnotes.
- This matters operationally because the planner never binds everything at once: the documented default `max\_selected\_tools` cap stays at 20, so breadth of capability does not mean uncontrolled tool sprawl per turn.

Kalier in v1.7.1

- Kalier remains the 67th workflow agent in `v1.7.1`, but its newest upgrade is usability: Tlmatini now acts as the embedded MCP-Kali-Server client for chat-side runs.
- It can issue one capability per run, including `nmap`, `gobuster`, `dirb`, `nikto`, `sqlmap`, `metasploit`, `hydra`, `john`, `wpscan`, `enum4linux`, arbitrary `command`, or a safe `health` probe of the remote server.
- The agent emits `INI\_SECTION\_KALIER` with `action`, `endpoint`, `subject`, `return\_code`, `success`, `timed\_out`, and `server\_url`, so downstream Forker or Parametrizer logic can branch on results without scraping prose.
- The practical operator result is simpler prompts: after the Kali box URL is configured once, normal Multi-Turn requests no longer repeat `server\_url` unless you intentionally override it for a one-off target.
- Two operator surfaces ship in lock-step: the wrapped Multi-Turn tool `chat\_agent\_kalier` now auto-injects the configured `kali\_server\_url`, while the visual Kalier canvas node still stores `server\_url` explicitly in YAML per node.
- Kalier talks straight to the Kali-side Flask API over HTTP using Python-stdlib `urllib`, so it stays self-contained in the pool subprocess and works the same in source or frozen builds; the embedded-client injection fails open if the config value is blank or unreadable.
- Authorized use only: the intended operator flow is an in-scope lab, CTF, or permitted engagement, often with `ssh -L 5000:localhost:5000 user@KALI\_IP` tunneling a remote Kali box back to `http://127.0.0.1:5000`, and the repo now includes `Tlmatini-Kali-Setup.md` for the zero-client walkthrough.

STM32er in v1.9.0

- STM32er is the new 68th workflow agent in `v1.9.0`: Tlmatini's bridge to the `STM32 Template Project MCP`, built so she can scaffold, author, build, flash, reset, and observe STM32F4 firmware without driving the STM32CubeIDE GUI.
- The operator promise is zero-config bootstrap: leave `server\_script` blank and STM32er downloads or zip-falls-back to the MCP server on first use, installs `mcp` and `pyserial` if needed, validates, and caches the result so the user only installs STM32CubeIDE plus Tlmatini.
- Before any compile-or-hardware action, STM32er runs a critical-mission fail-safe preflight over the arm-none-eabi toolchain, STM32CubeIDE, programmer path, ST-LINK probe, and target family; compile-only steps can run boardless, but flash, erase, reset, serial, and SWD/live-memory operations are refused when the environment or device is wrong.
- Two operator surfaces ship in lock-step: the wrapped Multi-Turn tool `chat\_agent\_stm32er` takes one `action` per call, while the visual STM32er canvas node stores the same fields in YAML and triggers downstream agents on both success and failure.
- The tool surface includes the full project lifecycle plus hardware-in-the-loop composites: `validate`, `bootstrap`, `create\_project`, `write\_source`, `build`, `build\_and\_flash`, `serial\_session`, `live\_monitor`, and the rest of the 23 MCP verbs, with every run emitting an `INI\_SECTION\_STM32ER` block for Forker or Parametrizer routing.

- Config -> URLs now seeds the chat path with ``stm32_mcp_server_script``, ``stm32_mcp_python``, ``stm32_template_dir``, ``stm32_ide_root``, ``stm32_mcp_repo_url``, and ``stm32_mcp_install_dir``, so firmware prompts normally describe only the task and target board instead of the plumbing.

Windower on Multi-Turn and canvas

- Windower is the new 66th workflow agent: a deterministic Win32 window manager that finds an application window by title and performs one lifecycle operation on the window itself instead of clicking inside it.
- It supports ``focus``, ``minimize``, ``maximize``, ``restore``, ``move``, ``resize``, ``move_resize``, ``close``, ``topmost``, ``untopmost``, ``arrange``, and ``list``, with ``substring``, ``exact``, or ``regex`` title matching plus ``match_index`` for duplicate titles.
- The agent emits ``INI_SECTION_WINDOWER`` with ``action``, ``window_title``, ``matched``, ``match_count``, ``state``, ``left``, ``top``, ``width``, and ``height``, so downstream Forker or Parametrizer logic can branch on presence, state, or geometry.
- Two operator surfaces ship in lock-step: the wrapped Multi-Turn tool ``chat_agent_windower`` accepts free-form key=value requests, while the visual Windower canvas node stores the same operation fields in YAML.
- Windower is the desktop-window sibling of Mouser and Keyboarder: use Windower when the goal is the window as a whole, Mouser for controls inside it, and Keyboarder for text entry into it.
- Because Windower changes real window state, its executions appear in Exec Report and it always triggers ``target_agents`` whether the action succeeds or fails.

Playwrighter in v1.5.0

- Playwrighter is the new 65th workflow agent in ``v1.5.0``: a deterministic Playwright-powered browser automator for Chromium, Firefox, or WebKit that executes ordered interactive steps instead of static fetches.
- It covers the gap between Crawler and Googler: logins, multi-step forms, wizard clicks, JS-rendered SPA scraping, screenshots after interaction, assertions, downloads, and authenticated end-to-end UI checks.
- The agent emits ``INI_SECTION_PLAYWRIGHTER`` with ``start_url``, ``final_url``, ``status``, ``steps_run``, ``assert_result``, and extracted values so downstream Forker or Parametrizer logic can branch on pass/fail or reuse scraped data.
- Two operator surfaces ship in lock-step: the wrapped Multi-Turn tool ``chat_agent_playwrighter`` takes the whole script as ``steps_json``, while the visual Playwrighter canvas node stores the same declarative step list in YAML.
- Session continuity is built in: ``headless: false`` lets operators watch it drive, and ``storage_state_in`` / ``storage_state_out`` carry login state across runs without forcing a manual browser setup each time.
- Because Playwrighter is state-changing, its executions appear in Exec Report and it always triggers ``target_agents`` whether the run succeeds or fails.

Reviewer precision patch in v1.4.1

- The ``v1.4.1`` Reviewer refinement tightened accuracy rather than adding a new surface: when ``diff_ref`` is empty, the review prompt labels the diff as the uncommitted working tree plus staged area, so the model must not describe those findings as already committed or pushed.
- The same patch teaches the model Tlmatini's managed-secret convention: ``agent/config.json`` and selected ``agent/agents/*/config.yaml`` files can hold live local credentials in a keyed working copy, while ``regen_secrets.py --mode push-able`` scrubs them back to placeholders before commit.
- That guidance is mirrored into the ``code-review`` SKILL.md package too, keeping the chat-surface review behavior and the canvas Reviewer agent aligned on commit-state wording and secret-severity expectations.

Native dialogs and Tkinter removal in v1.4.2

- The current tagged release ``v1.4.2`` removes Tkinter from the unstable runtime-facing dialog path and replaces it with ``Tlmatini/agent/native_dialogs.py``, a native Windows dialog bridge used by browser-triggered pickers.
- This change pairs with the existing DB and operator dialogs: file and folder selection still feels local and GUI-first, but the fragile Tkinter dependency is no longer part of the interactive runtime path that users trigger from chat or ACP surfaces.
- The patch arrived with dedicated tests (``test_native_dialogs.py``) and with follow-up orphan-reaper/runtime adjustments, so the release reads as a stability pass rather than a cosmetic refactor.

ACPX-Skills menu

- The new `ACPX-Skills` navbar menu gives operators four direct actions over the skill catalog: Browse Skills, Configure Skills, Diagnostics, and Reload Registry.
- `Configure Skills` flips `Skill.enabled` exactly the way MCPs and Tools are toggled, so disabled skills disappear from `list\_skills` and reject `invoke\_skill` with `SKILL\_DISABLED` instead of silently half-working.
- The diagnostics view cross-checks skill dependencies against disabled tools, disabled MCPs, missing ACPX agents, and orphan database rows whose SKILL.md disappeared from disk.

Source-mode bootstrap commands

```
python -m venv venv
venv\Scripts\activate
pip install -r requirements.txt
python Tlamatini/manage.py migrate
python Tlamatini/manage.py createsuperuser
python Tlamatini/manage.py collectstatic --noinput
python Tlamatini/manage.py runserver --noreload
```

4. Ollama Setup Without Administrative Rights

- Open a normal PowerShell window, not an elevated one, for the safest no-admin Windows installation path.
- Install into `%LOCALAPPDATA%\Programs\Ollama` with the official PowerShell installer script and then reopen PowerShell so PATH updates are visible.
- Verify the CLI with `ollama --version`, start `ollama serve` if the background service is not already active, and confirm `http://127.0.0.1:11434/api/tags` responds.
- Pull the default repository model tags exactly as written if you want the shipped config and agent templates to work unchanged.

No-admin Ollama commands and default model pulls

```
$env:OLLAMA_INSTALL_DIR = "$env:LOCALAPPDATA\Programs\ollama"
irm https://ollama.com/install.ps1 | iex
ollama --version
ollama serve
Invoke-WebRequest http://127.0.0.1:11434/api/tags -UseBasicParsing
ollama pull qwen3-embedding:8b
ollama pull glm-5:cloud
ollama pull qwen3.5:cloud
ollama pull gpt-oss:120b-cloud
ollama pull qwen3.5:397b-cloud
ollama pull llama3.2-vision:11b
```

5. Runtime, Release, and Operator Diagnostics

Running the application

- Development server: ``python Tlamatini/manage.py runserver --noreload``.
- Preferred async/dev bootstrap: ``python Tlamatini/manage.py startserver``, which starts MCP services before the Django server.
- Production-style ASGI endpoint: ``daphne -b 127.0.0.1 -p 8000 tlamatini.asgi:application``.
- Current startup also re-applies GPU-performance / Ollama-pinning hooks in the background on supported NVIDIA Windows hosts, so restart-time behavior stays closer to the tuned development baseline.
- That same early startup window is also where a staged ``DB/ToLoad/db.sqlite3`` file is promoted into the live database path, before Django opens SQLite.
- Startup now also prints a ``--- [VERSION] Tlamatini ...`` banner, making the running build visible in both the console and ``tlamatini.log`` without an HTTP call.
- Startup cleans pool state, repopulates the agent registry, launches MCP metrics/file-search servers, and then serves HTTP plus WebSocket traffic.

Embedding-memory pre-flight guard

- README and BookOfTlamatini now document an embedding-memory pre-flight guard for GPU hosts before a directory-context load starts its FAISS embedding burst.
- On supported NVIDIA hosts it estimates embedding-model VRAM pressure, and when the projected load is too high it emits a non-blocking warning chat bubble instead of silently letting RAM<->VRAM thrash surprise the operator.
- CPU-only, AMD, and Apple-Silicon hosts stay fail-open: the guard becomes a no-op when the NVIDIA probe does not apply.
- The practical operator response is straightforward: switch to a smaller embedding model, reconnect if needed, or proceed knowingly with the heavier model.

Reconnect and restart safeguards

- Config -> Models and Config -> URLs dialogs now track their pre-edit baseline and can show a reconnect-required dialog when the saved values change what the live chat session should trust.
- The restored-session autoload path now buffers early WebSocket frames so context-loading spinners and disabled-input state are not lost during automatic reconnect/restore flows.
- Startup and restart behavior now also re-apply GPU performance and Ollama keep-alive hooks in the background on supported NVIDIA Windows hosts, improving warm-model readiness without blocking Django boot.
- Windows process hygiene is now part of that safety story too: detached no-window spawns and the three-tier orphan reaper reduce the chance that Task Manager shows stale Tlamatini-icon console helpers after long runs.

Prompt catalog and answer readability discipline

- Version ``1.3.2`` tightened the HTML answer contract with a Prime Directive on visual readability: explicit background and text color, no grey-on-dark body text, and safer table-body defaults.
- The seeded ``Prompts`` dropdown was also re-sorted into a learner path: context-only Q&A first, then metrics, files search, shell, code generation, vision, specialized single-tool actions, agent control, Unrealer, and heavier Multi-Turn/ACPX demos last.
- Those readability rules remain in force in the current documentation set, and the newer ``v1.9.0`` release state keeps the version badge, runtime surfaces, self-knowledge wording, STM32er demo prompts, and operator handbook aligned.

Unreal MCP runtime behavior

- Each Unrealer run is one command: load ``config.yaml``, open a fresh TCP socket, send ``{"type": "command", "params": "params"}``, read until valid JSON arrives, and log one atomic ``INI_SECTION_UNREALER<<< ... >>>END_SECTION_UNREALER`` block.

- The wrapped-tool path stores chat runs under ``agent/agents/pools/_chat_runs/_unrealer_<seq>_<id>/'`, while visual flows use normal ``unrealer_<n>`` pool folders and can chain response fields through Parametrizer mappings.
- Exec Report treats Unrealer as its own row family, and troubleshooting stays concrete: connection-refused means the plugin is not listening, while read-timeout usually means UE5's game thread is busy.

Orphan reaper runtime behavior

- Tlamatini now ships a three-tier orphan reaper in ``Tlamatini/agent/orphan_reaper.py`` focused on Windows console-host leftovers such as ``conhost.exe`` and ``openconsole.exe``.
- Tier 1 runs after spawn-capable Multi-Turn tool calls, Tier 2 runs once after the final answer in a background thread, and Tier 3 runs again during application shutdown through the same cleanup path that already tears down pools.
- The candidate set stays narrow on purpose: dead descendants of the current process tree, orphaned console hosts tied to that tree, and pool-linked processes whose ``cmdline`` still points into ``agents/pools/...`` even though tracking records are gone.
- Prevention landed alongside cleanup: Windows spawn sites now use ``CREATE_NO_WINDOW | DETACHED_PROCESS | CREATE_NEW_PROCESS_GROUP`` with stdio piped to ``DEVNULL``, so the console host is usually never allocated in the first place.
- The ACPX runtime now kills full process trees instead of only top-level wrappers, and pool-agent scripts gained a conservative ``subprocess.Popen`` guard so forgotten descendants still inherit ``CREATE_NO_WINDOW``.
- If something survives both cleanup tiers, Tlamatini sends a second chat bubble listing each surviving ``name + PID``; the reaper never raises into the user path because a cleanup crash would be worse than the leftovers it tried to remove.

Release pipeline

- Release production is a three-step pipeline: ``build.py`` -> ``build_uninstaller.py`` -> ``build_installer.py``.
- The final distributable is the full ``dist/Tlamatini_Release/`` folder, not a stray executable copied outside its payload.
- Use ``build.py --self-modify`` when you intentionally want a release that ships ``TlamatiniSourceCode/`` so she can inspect or modify herself at runtime.
- Current ``build.py`` treats ``README.md`` and ``jd-cli/`` as required post-build assets and fails hard if those payloads are missing.
- Bundled support scripts cover shortcut creation/removal, ``.flw`` association, the PowerShell launcher, and Windows-specific installer ergonomics.

Exec Report

- Exec Report is a Multi-Turn-only transparency layer that appends one operation table per state-changing agent family to the final answer.
- Rows are recorded from the live tool-call stream rather than guessed from the LLM prose, so the report is the operational ground truth.
- Each row receives a SUCCESS/FAILURE verdict from raw tool returns, making long installs, deployments, and remediations inspectable after the fact.

ACPX and skills

- ACPX lets Tlamatini spawn external coding-agent CLIs such as Codex, Claude Code, Cursor, Gemini, Qwen, and others as managed child processes.
- It pairs those agents with markdown-driven ``SKILL.md`` packages, validated I/O contracts, permission gating, and append-only audit logs.
- Transport-aware drain rules and bounded event bodies reduce latency while protecting the LLM context budget during external-agent relays.
- The operator-facing references are ``README.md`` and ``ACPX.md``, while the implementation lives under ``agent/acpx/``, ``agent/skills/``, and ``agent/skills_pkg/``.

Application log

- The built-in ``tlatini.log`` file captures both stdout and stderr through a tee stream initialized in ``manage.py`` before Django starts.
- In source mode the log sits next to ``manage.py``; in frozen mode it lives next to the executable.
- Immediate flush behavior makes the log the primary forensic artifact for startup problems, warnings, tracebacks, and runtime diagnostics.

6. Agent Catalog and Runtime Model

Tlmatini currently exposes 68 workflow-agent templates.

Category	Representative agents
Control	starter, ender, stopper, cleaner, barrier, flowbacker
Execution and files	executer, pythonxer, pser, file_creator, file_extractor, file_interpreter, de_compressor, playwrighter, windower, unrealer
DevOps and infra	gitter, dockerer, kubernetes, jenkinser, ssher, scper
Data and APIs	sqler, mongoxer, apirer, crawler, googler
Monitoring and routing	monitor_log, monitor_netstat, flowhypervisor, forker, asker, counter, and, or
Communication	notifier, emailer, recmailer, telegramer, telegramrx, teletlmatini, whatsapper, whatstlmatini
Security and media	kyber_keygen, kyber_cipher, kyber_decipher, image_interpreter, shoter, j_decompiler
Workflow intelligence	flowcreator, gatewayer, gateway_relayer, node_manager, parametrizer, prompter, summarizer, acpxer

All workflow agents follow a common deployment pattern: template directory, YAML configuration, session-scoped pool copy, PID/status/log files, target/source wiring, and optional reanimation state.

Agent catalog validation

- The authoritative human-readable source for workflow-agent descriptions is `agents\_descriptions.md` at the repo root, not an embedded JavaScript map or a hard-coded Django list.
- Current validation compares live template directories, `agents\_descriptions.md` rows, and the README / Book bestiary counts so the generated dossier stays aligned with the running product.
- The ACP sidebar hover tooltip and the right-click Description dialog both resolve through that markdown file first; `README.md` is only a legacy fallback when the dedicated description file is absent.

How agent runtimes are shaped

- Every workflow agent follows the same operational skeleton: template directory, `config.yaml`, a session-scoped pool copy, PID/status/log files, and explicit source/target wiring.
- Chat-wrapped tool calls launch isolated runtime copies under `agent/agents/pools/\_chat\_runs\_/\_`, while ACP uses named pool folders such as `starter\_1` or `unrealer\_1`.
- Specialized agents now stretch the platform in different directions: ACPXer drives external coding-agent CLIs, Kalier drives a remote or tunneled Kali Linux tool server, STM32er drives a zero-config STM32 firmware MCP bridge, Unrealer drives a live UE5 editor, and TeleTlmatini / WhatsTlmatini bridge full Tlmatini conversations into messaging platforms.

Reviewer and Analyzer spotlight

- The workflow catalog now includes two new high-value specialists: Reviewer and Analyzer, lifting the canvas inventory to 64 agents and the seed-skill catalog to 23 SKILL.md packages.
- Reviewer is LLM-powered: it resolves a git diff for `repo\_path`, reviews it with a senior-engineer prompt, and emits an `INI\_SECTION\_REVIEWER` block whose first routable field is `verdict = APPROVE | REQUEST\_CHANGES | COMMENT`.
- Analyzer is deterministic and scanner-driven: it runs whichever of `bandit`, `semgrep`, `ruff`, `eslint`, `gitleaks`, and `pip-audit` are installed on PATH over `target\_path`, then emits an `INI\_SECTION\_ANALYZER` block with `status` and `total\_findings` for downstream routing.
- Both agents always trigger `target\_agents`, so a downstream Forker can branch on `{verdict}`, `{status}`, or `{total\_findings}` instead of scraping prose.
- They are intentionally canvas-only workflow agents: there is no wrapped `chat\_agent\_reviewer` or `chat\_agent\_analyzer`, and therefore no duplicate Exec Report row family for those names.
- The chat-side counterparts live in the ACPX skill catalog instead: `code-review` exposes senior-engineer git-diff review, while `security-audit` exposes the deterministic multi-scanner sweep.

Operator surface counts

- The live operator surface now stands at 68 workflow agents, 75 Multi-Turn tools, 12 ACPX tools, and 24 skills.
- Source inspection confirms the newer total: 43 wrapped chat-agent tools in ``chat_agent_registry.py``, which combines with 20 core Python tools and 12 ACPX/Skill tools for 75 Multi-Turn tools overall.
- README.md and BookOfTlmatini.md now agree on those counts, so the dossier can mirror the docs and the live registry without reconciliation footnotes.
- This matters operationally because the planner never binds everything at once: the documented default ``max_selected_tools`` cap stays at 20, so breadth of capability does not mean uncontrolled tool sprawl per turn.

Kalier spotlight

- Kalier remains the 67th workflow agent in ``v1.7.1``, but its newest upgrade is usability: Tlmatini now acts as the embedded MCP-Kali-Server client for chat-side runs.
- It can issue one capability per run, including ``nmap``, ``gobuster``, ``dirb``, ``nikto``, ``sqlmap``, ``metasploit``, ``hydra``, ``john``, ``wpscan``, ``enum4linux``, arbitrary ``command``, or a safe ``health`` probe of the remote server.
- The agent emits ``INI_SECTION_KALIER`` with ``action``, ``endpoint``, ``subject``, ``return_code``, ``success``, ``timed_out``, and ``server_url``, so downstream Forker or Parametrizer logic can branch on results without scraping prose.
- The practical operator result is simpler prompts: after the Kali box URL is configured once, normal Multi-Turn requests no longer repeat ``server_url`` unless you intentionally override it for a one-off target.
- Two operator surfaces ship in lock-step: the wrapped Multi-Turn tool ``chat_agent_kalier`` now auto-injects the configured ``kali_server_url``, while the visual Kalier canvas node still stores ``server_url`` explicitly in YAML per node.
- Kalier talks straight to the Kali-side Flask API over HTTP using Python-stdlib ``urllib``, so it stays self-contained in the pool subprocess and works the same in source or frozen builds; the embedded-client injection fails open if the config value is blank or unreadable.
- Authorized use only: the intended operator flow is an in-scope lab, CTF, or permitted engagement, often with ``ssh -L 5000:localhost:5000 user@KALI_IP`` tunneling a remote Kali box back to ``http://127.0.0.1:5000``, and the repo now includes ``Tlmatini-Kali-Setup.md`` for the zero-client walkthrough.

STM32er spotlight

- STM32er is the new 68th workflow agent in ``v1.9.0``: Tlmatini's bridge to the ``STM32 Template Project MCP``, built so she can scaffold, author, build, flash, reset, and observe STM32F4 firmware without driving the STM32CubeIDE GUI.
- The operator promise is zero-config bootstrap: leave ``server_script`` blank and STM32er downloads or zip-falls-back to the MCP server on first use, installs ``mcp`` and ``pyserial`` if needed, validates, and caches the result so the user only installs STM32CubeIDE plus Tlmatini.
- Before any compile-or-hardware action, STM32er runs a critical-mission fail-safe preflight over the arm-none-eabi toolchain, STM32CubeIDE, programmer path, ST-LINK probe, and target family; compile-only steps can run boardless, but flash, erase, reset, serial, and SWD/live-memory operations are refused when the environment or device is wrong.
- Two operator surfaces ship in lock-step: the wrapped Multi-Turn tool ``chat_agent_stm32er`` takes one ``action`` per call, while the visual STM32er canvas node stores the same fields in YAML and triggers downstream agents on both success and failure.
- The tool surface includes the full project lifecycle plus hardware-in-the-loop composites: ``validate``, ``bootstrap``, ``create_project``, ``write_source``, ``build``, ``build_and_flash``, ``serial_session``, ``live_monitor``, and the rest of the 23 MCP verbs, with every run emitting an ``INI_SECTION_STM32ER`` block for Forker or Parametrizer routing.
- Config -> URLs now seeds the chat path with ``stm32_mcp_server_script``, ``stm32_mcp_python``, ``stm32_template_dir``, ``stm32_ide_root``, ``stm32_mcp_repo_url``, and ``stm32_mcp_install_dir``, so firmware prompts normally describe only the task and target board instead of the plumbing.

Windower spotlight

- Windower is the new 66th workflow agent: a deterministic Win32 window manager that finds an application window by title and performs one lifecycle operation on the window itself instead of clicking inside it.

- It supports ``focus``, ``minimize``, ``maximize``, ``restore``, ``move``, ``resize``, ``move_resize``, ``close``, ``topmost``, ``untopmost``, ``arrange``, and ``list``, with ``substring``, ``exact``, or ``regex`` title matching plus ``match_index`` for duplicate titles.
- The agent emits ``INI_SECTION_WINDOWER`` with ``action``, ``window_title``, ``matched``, ``match_count``, ``state``, ``left``, ``top``, ``width``, and ``height``, so downstream Forker or Parametrizer logic can branch on presence, state, or geometry.
- Two operator surfaces ship in lock-step: the wrapped Multi-Turn tool ``chat_agent_windower`` accepts free-form key=value requests, while the visual Windower canvas node stores the same operation fields in YAML.
- Windower is the desktop-window sibling of Mouser and Keyboarder: use Windower when the goal is the window as a whole, Mouser for controls inside it, and Keyboarder for text entry into it.
- Because Windower changes real window state, its executions appear in Exec Report and it always triggers ``target_agents`` whether the action succeeds or fails.

Playwrighter spotlight

- Playwrighter is the new 65th workflow agent in ``v1.5.0``: a deterministic Playwright-powered browser automator for Chromium, Firefox, or WebKit that executes ordered interactive steps instead of static fetches.
- It covers the gap between Crawler and Googler: logins, multi-step forms, wizard clicks, JS-rendered SPA scraping, screenshots after interaction, assertions, downloads, and authenticated end-to-end UI checks.
- The agent emits ``INI_SECTION_PLAYWRIGHTER`` with ``start_url``, ``final_url``, ``status``, ``steps_run``, ``assert_result``, and extracted values so downstream Forker or Parametrizer logic can branch on pass/fail or reuse scraped data.
- Two operator surfaces ship in lock-step: the wrapped Multi-Turn tool ``chat_agent_playwrighter`` takes the whole script as ``steps_json``, while the visual Playwrighter canvas node stores the same declarative step list in YAML.
- Session continuity is built in: ``headless: false`` lets operators watch it drive, and ``storage_state_in`` / ``storage_state_out`` carry login state across runs without forcing a manual browser setup each time.
- Because Playwrighter is state-changing, its executions appear in Exec Report and it always triggers ``target_agents`` whether the run succeeds or fails.

Reviewer precision spotlight

- The ``v1.4.1`` Reviewer refinement tightened accuracy rather than adding a new surface: when ``diff_ref`` is empty, the review prompt labels the diff as the uncommitted working tree plus staged area, so the model must not describe those findings as already committed or pushed.
- The same patch teaches the model Tlmatini's managed-secret convention: ``agent/config.json`` and selected ``agent/agents/*/config.yaml`` files can hold live local credentials in a keyed working copy, while ``regen_secrets.py --mode push-able`` scrubs them back to placeholders before commit.
- That guidance is mirrored into the ``code-review`` SKILL.md package too, keeping the chat-surface review behavior and the canvas Reviewer agent aligned on commit-state wording and secret-severity expectations.

Native-dialog spotlight

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- This change pairs with the existing DB and operator dialogs: file and folder selection still feels local and GUI-first, but the fragile Tkinter dependency is no longer part of the interactive runtime path that users trigger from chat or ACP surfaces.
- The patch arrived with dedicated tests (``test_native_dialogs.py``) and with follow-up orphan-reaper/runtime adjustments, so the release reads as a stability pass rather than a cosmetic refactor.

Unrealer spotlight

- Unreal MCP is a UE5 plugin that runs inside the editor and listens on ``127.0.0.1:55557`` for one JSON command per TCP connection; Tlmatini is the client side of that link, not the plugin host.
- The Unrealer workflow agent forwards any command the connected plugin build exposes, up to a 53-command surface across nine categories: actor manipulation (incl. viewport screenshots), Blueprint authoring and node wiring, input mappings, UMG widget building, in-editor Python/console execution, level I/O, asset import, and material authoring. Headless build/cook/test is out of scope (it needs UnrealEditor-Cmd, not the editor socket).

- The same integration is available in both operator surfaces: checked Multi-Turn via ``chat_agent_unrealer``, and ACP canvas flows via the visual Unrealer node plus Parametrizer chaining.
- Each Unrealer run is one command: load ``config.yaml``, open a fresh TCP socket, send ``{"type": "command", "params": params}``, read until valid JSON arrives, and log one atomic ``INI_SECTION_UNREALER<<< ... >>>END_SECTION_UNREALER`` block.
- The wrapped-tool path stores chat runs under ``agent/agents/pools/_chat_runs/_unrealer_<seq>_<id>/``, while visual flows use normal ``unrealer_<n>`` pool folders and can chain response fields through Parametrizer mappings.
- Exec Report treats Unrealer as its own row family, and troubleshooting stays concrete: connection-refused means the plugin is not listening, while read-timeout usually means UE5's game thread is busy.

De-Compressor spotlight

- De-Compressor is the deterministic archive worker for compression and decompression tasks, deciding direction from whichever side exposes a recognized archive extension.
- Supported archive families are ``.gz``, ``.zip``, ``.7z``, ``.tar.gz``, and ``.gz.tar``; file-to-folder extraction and file-or-directory packing are both documented in the README and Book.
- Password handling is explicit: ``passwordless=true`` skips it, while ``passwordless=false`` requires the ``DE_COMPRESSER_PWD`` environment variable and fails fast if the secret is missing.
- The agent is reachable from both operator surfaces: ACP canvas nodes wire through dedicated connection-update views, and checked Multi-Turn can invoke it through ``chat_agent_de_compressor``.
- Format engines are practical rather than magical: `stdlib`gzip`` and ``zipfile`` cover core cases, ``.7z`` is preferred for encrypted ``.7z`` or ``.zip``, and ``py7zr`` was added to ``requirements.txt`` as the Python fallback.
- Every run emits an ``INI_SECTION_DE_COMPRESSER<<< ... >>>END_SECTION_DE_COMPRESSER`` block and still triggers ``target_agents``, so downstream Parametrizer or Raiser logic can branch on ``success=true|false`` instead of guessing from prose.

Orphan-reaper spotlight

- Tlmatini now ships a three-tier orphan reaper in ``Tlmatini/agent/orphan_reaper.py`` focused on Windows console-host leftovers such as ``conhost.exe`` and ``openconsole.exe``.
- Tier 1 runs after spawn-capable Multi-Turn tool calls, Tier 2 runs once after the final answer in a background thread, and Tier 3 runs again during application shutdown through the same cleanup path that already tears down pools.
- The candidate set stays narrow on purpose: dead descendants of the current process tree, orphaned console hosts tied to that tree, and pool-linked processes whose ``cmdline`` still points into ``agents/pools/...`` even though tracking records are gone.

7. Repository Facts and Git Changes

Metric	Value
Tracked files in git	505
Workflow agents	68
Multi-Turn tools	75
Wrapped chat-agent tools	43
Skills	24
agents_descriptions.md rows	68
Django migrations	103
Frontend JavaScript modules	28
Frontend CSS files	6
HTML templates	4
Python requirements	71
Binary/asset tracked files skipped from line count	22
Resolved version	1.9.0
Version source	git

Latest commits

Date	Commit	Subject
2026-05-26	b9ad650	Implementing STMer agent, now Tlmatini is able to program completely in hardware of STM32F processors (HARDWARE), by and
2026-05-26	784e425	Another video of me of magic XD, by Tlmatini's-AutoBot.
2026-05-26	8c8e290	Another image of me, by Tlmatini's-AutoBot.
2026-05-26	be5517c	A video of me welcoming, by Tlmatini's-AutoBot.
2026-05-26	2dd65e7	Just a photo of myself, by Tlmatini's-AutoBot.
2026-05-25	af17c2c	Remaning the image of me waving my hand, by Tlmatini's-AutoBot.
2026-05-25	26d85ba	Another image of me, by TTlmatini's-AutoBot.
2026-05-25	f28800f	Added some pics of mine in person, by Tlmatini's-AutoBot.
2026-05-25	d8bef06	Setting of version 1.8.0 in all of the documentation, by Tlmatini's-AutoBot.
2026-05-25	95cf6e2	Tlmatini, now She knows about herself, now enabled to modify herself, by Tlmatini's-AutoBot.

Changes since the last committed PDF/PPTX refresh

Last committed visual-dossier refresh: d8bef06 on 2026-05-25 — Setting of version 1.8.0 in all of the documentation, by Tlmatini's-AutoBot.

- Since the last committed PDF/PPTX refresh, `v1.9.0` added STM32er as the 68th workflow agent and `chat\_agent\_stm32er` as the new firmware-control surface: she now bridges the `STM32 Template Project MCP` to scaffold, build, flash, observe, and reset STM32F4 projects with a fail-safe hardware preflight.

Visual-dossier change appendix 1 of 1

Date	Commit	Subject
2026-05-26	b9ad650	Implementing STMer agent, now Tlmatini is able to program completely in hardware of STM32F processors (HARDWARE), by and
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2026-05-25	26d85ba	Another image of me, by TTlamatini's-AutoBot.
2026-05-25	f28800f	Added some pics of mine in person, by Tlamatini's-AutoBot.

Git changes from today

- Today's headline change is `v1.9.0` STM32er: Tlamatini now reaches 68 workflow agents and adds a zero-config bridge into the `STM32 Template Project MCP`, so she can scaffold, build, flash, observe, and reset STM32F4 firmware from chat or canvas while a critical-mission safety preflight refuses the wrong toolchain, probe, or device family.

Today's commit appendix 1 of 1

Date	Commit	Subject
2026-05-26	b9ad650	Implementing STMer agent, now Tlamatini is able to program completely in hardware of STM32F processors (HARDWARE), by an
2026-05-26	784e425	Another video of me of magic XD, by Tlamatini's-AutoBot.
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2026-05-26	2dd65e7	Just a photo of myself, by Tlamatini's-AutoBot.

8. Effective Line Inventory by Language

Methodology: tracked text files only. Blank lines and comment-only lines are excluded. Python counts also remove module, class, and function docstrings detected through AST parsing.

Language	Files	Effective lines	Total lines	Share
Python	292	76,477	106,863	68.3%
Markdown	58	13,454	17,206	12.0%
JavaScript	28	13,204	16,612	11.8%
CSS	6	3,431	4,347	3.1%
YAML	72	1,529	3,301	1.4%
HTML	4	829	847	0.7%
Other text	3	789	967	0.7%
Prompt template	2	678	793	0.6%
PowerShell	5	506	752	0.5%
JavaScript module	1	393	460	0.4%
JSON	5	392	392	0.4%
Text	5	183	213	0.2%
Protocol Buffers	1	20	33	0.0%
Batch	1	13	24	0.0%

Total effective lines: 111,898

9. Largest Effective Source Files

Path	Language	Effective	Total
Tlmatini/agent/views.py	Python	7,289	10,560
Tlmatini/agent/tests.py	Python	4,186	5,375
Tlmatini/agent/tools.py	Python	2,239	3,184
Tlmatini/agent/doc_generation/complete_project_docs.py	Python	2,018	2,269
Tlmatini/agent/agents/flowcreator/agent_skill.md	Markdown	1,963	2,176
BookOfTlmatini.md	Markdown	1,930	2,724
Tlmatini/agent/static/agent/css/agent_control_panel.css	CSS	1,720	2,298
Tlmatini/agent/static/agent/js/agent_page_init.js	JavaScript	1,415	1,655
Tlmatini/agent/static/agent/js/acp-canvas-core.js	JavaScript	1,303	1,613
README.md	Markdown	1,240	1,760
Tlmatini/agent/consumers.py	Python	1,240	1,545
Tlmatini/agent/static/agent/js/acp-agent-connectors.js	JavaScript	1,217	1,363
Tlmatini/agent/test_stm32er_agent.py	Python	1,200	1,707
Tlmatini/agent/agents/stm32er/stm32er.py	Python	1,191	1,642
Tlmatini/agent/static/agent/css/agent_page.css	CSS	1,181	1,433
Tlmatini/agent/static/agent/js/agent_page_chat.js	JavaScript	1,148	1,574
Tlmatini/agent/agents/whatstlmatini/whatstlmatini.py	Python	1,059	1,323
Tlmatini/agent/static/agent/js/acp-control-buttons.js	JavaScript	1,036	1,344
Tlmatini/agent/acpx/tests.py	Python	1,035	1,299
KIMI.md	Markdown	1,004	1,218
Tlmatini/agent/chat_agent_registry.py	Python	923	942
Tlmatini/agent/agents/teletlmatini/teletlmatini.py	Python	903	1,170
Tlmatini/agent/static/agent/js/canvas_item_dialog.js	JavaScript	898	1,165
Tlmatini/agent/agents/crawler/crawler.py	Python	896	1,285
Tlmatini/agent/static/agent/js/acp-canvas-undo.js	JavaScript	883	994

10. Complete Tracked File Tree (Repository Appendix)

Tree appendix 1 of 9

```
Tlamatini/
|-- .gitignore
|-- _acpx_pipeline_demo_report.md
|-- ACPX.md
|-- agents_descriptions.md
|-- BookOfTlamatini.md
|-- build.py
|-- build_installer.py
|-- build_uninstaller.py
|-- Claude-Desktop-KALI-MCP-Session.md
|-- CLAUDE.md
|-- CreateShortcut.json
|-- CreateShortcut.ps1
|-- eslint.config.mjs
|-- example_of_prompt_to_make_de-compressor_agent.txt
|-- install.py
|-- KIMI.md
|-- LICENSE
|-- OllamaPricing.png
|-- package.json
|-- pnpm-lock.yaml
|-- prompt_example_2_create_agent.txt
|-- README.md
|-- regen_secrets.py
|-- register_flw.ps1
|-- RemoveShortcut.ps1
|-- requirements.txt
|-- Tlamatini-Kali-Setup.md
|-- Tlamatini.ico
|-- Tlamatini.jpg
|-- Tlamatini.ps1
|-- tlamatini_app_summary.pdf
|-- Tlamatini_extended_Artificial_Intelligence_Humanly_Tempered.pptx
|-- uninstall.py
|-- unregister_flw.ps1
|-- VERSIONING.md
|-- versioning.py
|-- .codex/
|   |-- skills/
|   |   |-- full-project-pdf-dossier/
|   |   |   |-- SKILL.md
|   |   |-- overlap-safe-pptx-dossier/
|   |   |   |-- SKILL.md
|   |-- .github/
|   |   |-- workflows/
|   |   |   |-- name-guard.yml
|   |-- docs/
|   |   |-- claude/
|   |   |   |-- acpx.md
|   |   |   |-- agents.md
|   |   |   |-- architecture.md
|   |   |   |-- exec-report.md
|   |   |   |-- frontend.md
|   |   |   |-- gotchas.md
|   |   |   |-- INDEX.md
|   |   |   |-- mcp-tools.md
|   |   |   |-- multi-turn.md
|   |   |   |-- recent-fixes.md
|   |-- pyinstaller_hooks/
|   |   |-- hook-numpy.py
|   |-- Tlamatini/
|   |   |-- cat_art.py
|   |   |-- manage.py
|   |   |-- .agents/
|   |   |   |-- workflows/
|   |   |   |   |-- create_new_agent.md
|   |   |-- .mcps/
|   |   |   |-- create_new_mcp.md
|   |   |-- agent/
|   |   |   |-- __init__.py
|   |   |   |-- admin.py
|   |   |   |-- apps.py
|   |   |   |-- capability_registry.py
|   |   |   |-- chain_files_search_lcel.py
|   |   |   |-- chain_system_lcel.py
|   |   |   |-- chat_agent_registry.py
```

Tree appendix 2 of 9

```
-- chat_agent_runtime.py
-- chat_history_loader.py
-- command_togen_pb2.txt
-- config.json
-- config_loader.py
-- constants.py
-- consumers.py
-- embedding_memory_guard.py
-- filesearch.proto
-- filesearch_pb2.py
-- filesearch_pb2_grpc.py
-- global_execution_planner.py
-- global_state.py
-- gpu_perf.py
-- inet_determiner.py
-- mcp_agent.py
-- mcp_files_search_client.py
-- mcp_files_search_server.py
-- mcp_system_client.py
-- mcp_system_server.py
-- models.py
-- native_dialogs.py
-- orphan_reaper.py
-- path_guard.py
-- prompt.pmt
-- rag_enhancements.py
-- routing.py
-- test_build_scripts.py
-- test_de_compressor.py
-- test_embedding_memory_guard.py
-- test_flow_contracts.py
-- test_kalier_agent.py
-- test_native_dialogs.py
-- test_orphan_reaper.py
-- test_password_quoting.py
-- test_playwright_agent.py
-- test_stm32er_agent.py
-- test_unrealer_agent.py
-- test_windower_agent.py
-- tests.py
-- Tlamatini.md
-- tools.py
-- urls.py
-- version.py
-- views.py
-- web_search_llm.py
-- acpx/
|  -- __init__.py
|  -- agent_registry.py
|  -- config.py
|  -- permissions.py
|  -- runtime.py
|  -- service.py
|  -- session_store.py
|  -- tests.py
|  -- tools.py
|  -- windows_spawn.py
-- agents/
|  -- acpxer/
|  |  -- acpxer.py
|  |  -- config.yaml
|  -- analyzer/
|  |  -- analyzer.py
|  |  -- config.yaml
|  -- and/
|  |  -- and.py
|  |  -- config.yaml
|  -- apirer/
|  |  -- apirer.py
|  |  -- config.yaml
|  -- asker/
|  |  -- asker.py
|  |  -- config.yaml
|  -- barrier/
|  |  -- barrier.py
|  |  -- config.yaml
```

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```
-- test_barrier.py
-- cleaner/
|   |-- __init__.py
|   |-- cleaner.py
|   |-- config.yaml
-- counter/
|   |-- config.yaml
|   |-- counter.py
-- crawler/
|   |-- config.yaml
|   |-- crawler.py
-- croner/
|   |-- config.yaml
|   |-- croner.py
-- de_compressor/
|   |-- config.yaml
|   |-- de_compressor.py
-- deleter/
|   |-- config.yaml
|   |-- deleter.py
-- dockerer/
|   |-- config.yaml
|   |-- dockerer.py
-- emailer/
|   |-- config.yaml
|   |-- emailer.py
-- ender/
|   |-- config.yaml
|   |-- ender.py
-- executer/
|   |-- config.yaml
|   |-- executer.py
-- file_creator/
|   |-- config.yaml
|   |-- file_creator.py
-- file_extractor/
|   |-- config.yaml
|   |-- file_extractor.py
-- file_interpreter/
|   |-- config.yaml
|   |-- file_interpreter.py
-- flowbacker/
|   |-- config.yaml
|   |-- flowbacker.py
-- flowcreator/
|   |-- agentic_skill.md
|   |-- config.yaml
|   |-- flowcreator.py
-- flowhypervisor/
|   |-- config.yaml
|   |-- flowhypervisor.py
|   |-- hypervisor_alert.wav
|   |-- monitoring-prompt.pmt
-- forker/
|   |-- config.yaml
|   |-- forker.py
-- gateway_relayer/
|   |-- config.yaml
|   |-- gateway_relayer.md
|   |-- gateway_relayer.py
-- gatewayer/
|   |-- config.yaml
|   |-- gatewayer.md
|   |-- gatewayer.py
|   |-- gatewayer_explanation.md
-- gitter/
|   |-- config.yaml
|   |-- gitter.py
-- googler/
|   |-- config.yaml
|   |-- googler.py
-- image_interpreter/
|   |-- config.yaml
|   |-- image_interpreter.py
-- j_decompiler/
|   |-- config.yaml
```

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```
| | | `-- j_decompiler.py
| | | -- jenkinser/
| | | | -- config.yaml
| | | | `-- jenkinser.py
| | | -- kalier/
| | | | -- config.yaml
| | | | `-- kalier.py
| | | -- keyboarder/
| | | | -- config.yaml
| | | | `-- keyboarder.py
| | | -- kubernetes/
| | | | -- config.yaml
| | | | `-- kubernetes.py
| | | -- kyber_cipher/
| | | | -- config.yaml
| | | | `-- kyber_cipher.py
| | | -- kyber_decipher/
| | | | -- config.yaml
| | | | `-- kyber_decipher.py
| | | -- kyber_keygen/
| | | | -- config.yaml
| | | | `-- kyber_keygen.py
| | | -- mongoxer/
| | | | -- config.yaml
| | | | `-- mongoxer.py
| | | -- monitor_log/
| | | | -- config.yaml
| | | | `-- monitor_log.py
| | | -- monitor_netstat/
| | | | -- config.yaml
| | | | `-- monitor_netstat.py
| | | -- mouser/
| | | | -- config.yaml
| | | | `-- mouser.py
| | | -- mover/
| | | | -- config.yaml
| | | | `-- mover.py
| | | -- node_manager/
| | | | -- config.yaml
| | | | -- node_manager.md
| | | | -- node_manager.py
| | | | -- nodes_example.json
| | | | `-- nodes_example.yaml
| | | -- notifier/
| | | | -- config.yaml
| | | | `-- notifier.py
| | | -- or/
| | | | -- config.yaml
| | | | `-- or.py
| | | -- parametrizer/
| | | | -- config.yaml
| | | | `-- parametrizer.py
| | | -- playwrighter/
| | | | -- config.yaml
| | | | `-- playwrighter.py
| | | -- prompter/
| | | | -- config.yaml
| | | | `-- prompter.py
| | | -- pser/
| | | | -- config.yaml
| | | | `-- pser.py
| | | -- pythonxer/
| | | | -- config.yaml
| | | | `-- pythonxer.py
| | | -- raiser/
| | | | -- config.yaml
| | | | `-- raiser.py
| | | -- recmailer/
| | | | -- config.yaml
| | | | `-- recmailer.py
| | | -- reviewer/
| | | | -- config.yaml
| | | | `-- reviewer.py
| | | -- scper/
| | | | -- config.yaml
| | | | `-- scper.py
```

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```
-- shooter/
| |-- config.yaml
| |-- shooter.py
-- sleeper/
| |-- config.yaml
| |-- sleeper.py
-- sqler/
| |-- config.yaml
| |-- sqler.py
-- ssher/
| |-- config.yaml
| |-- ssher.py
-- starter/
| |-- config.yaml
| |-- starter.py
-- stm32er/
| |-- config.yaml
| |-- stm32er.py
-- stopper/
| |-- config.yaml
| |-- stopper.py
-- summarizer/
| |-- config.yaml
| |-- summarizer.py
-- telegramer/
| |-- config.yaml
| |-- telegramer.py
-- telegramrx/
| |-- config.yaml
| |-- telegramrx.py
-- teletlamatini/
| |-- config.yaml
| |-- teletlamatini.py
-- unrealer/
| |-- config.yaml
| |-- unrealer.py
-- whatsapper/
| |-- config.yaml
| |-- whatsapper.py
-- whatstlamatini/
| |-- config.yaml
| |-- whatstlamatini.py
-- windower/
| |-- config.yaml
| |-- windower.py
-- doc_generation/
| |-- complete_project_docs.py
| |-- markdown_to_pdf.py
| |-- refresh_project_docs.py
| |-- test_output.pdf
-- images/
| |-- HiIamTlamatini.png
| |-- mockup.jpg
| |-- RealisticTlamatini.png
| |-- ShouldersUoTlamatini.png
| |-- Tlamatini AI Agentic Advisor Video.mp4
| |-- TlamatiniAbout.jpg
| |-- TlamatiniDrawings.png
| |-- TlammatiniLetsMakeSomeMagic.mp4
| |-- TorsoTlamatini.png
| |-- XAIHT-Tlamatini.mp4
-- imaging/
| |-- converter.py
| |-- image_interpreter.py
| |-- screenshot.py
-- management/
| |-- __init__.py
| |-- commands/
| | |-- acpx_lint.py
| | |-- startserver.py
-- migrations/
| |-- 0001_initial.py
| |-- 0002_populate_db.py
| |-- 0003_add_cleaner.py
| |-- 0004_add_deleter.py
| |-- 0005_add_executor.py
```

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```
-- 0006_add_notifier.py
-- 0007_add_stopper.py
-- 0008_add_recmailer.py
-- 0009_add_whatsapp.py
-- 0010_add_pythonxer.py
-- 0011_add_asker.py
-- 0012_repopulate_all_agents.py
-- 0013_add_forker.py
-- 0014_repopulate_all_agents.py
-- 0015_add_shooter.py
-- 0016_repopulate_all_agents.py
-- 0017_add_ssh.py
-- 0018_repopulate_all_agents.py
-- 0019_add_scanner.py
-- 0020_repopulate_all_agents.py
-- 0021_add_telegram.py
-- 0022_repopulate_all_agents.py
-- 0023_add_mongoxer.py
-- 0024_repopulate_all_agents.py
-- 0025_add_telegramer.py
-- 0026_repopulate_all_agents.py
-- 0027_add_sql.py
-- 0028_repopulate_all_agents.py
-- 0029_add_prompter.py
-- 0030_repopulate_all_agents.py
-- 0031_add_flowcreator.py
-- 0032_repopulate_all_agents.py
-- 0033_add_gitter.py
-- 0034_add_docker.py
-- 0035_add_ps.py
-- 0036_add_kubernetes.py
-- 0037_add_api.py
-- 0038_add_jenkins.py
-- 0039_add_crawler.py
-- 0040_add_summarizer.py
-- 0041_add_flowhypervisor.py
-- 0042_add_mouse.py
-- 0043_agentmessage_conversation_user.py
-- 0044_add_counter.py
-- 0045_add_file_interpreter.py
-- 0046_add_image_interpreter.py
-- 0047_add_gateway.py
-- 0048_add_gateway_relayer.py
-- 0049_add_node_manager.py
-- 0050_add_file_creator.py
-- 0051_add_file_extractor.py
-- 0052_add_kyber_keygen.py
-- 0053_add_kyber_cipher.py
-- 0054_add_kyber_decipher.py
-- 0055_fix_kyber_cipher_names.py
-- 0056_add_parametrizer.py
-- 0057_add_flowbacker.py
-- 0058_add_barrier.py
-- 0059_add_j_decompiler.py
-- 0060_add_agent_parametrizer_tool.py
-- 0061_add_agent_starter_tool.py
-- 0062_sync_agent_control_tools_and_prompts.py
-- 0063_add_agent_parametrizer_prompt.py
-- 0064_add_chat_agent_run_model.py
-- 0065_add_chat_wrapped_agent_tools.py
-- 0066_add_keyboard.py
-- 0067_add_multi_turn_demo_prompts.py
-- 0068_add_googler_tool.py
-- 0069_add_googler_agent.py
-- 0070_add_tellamini.py
-- 0071_acpx_skills.py
-- 0072_add_acpx_demo_prompts.py
-- 0073_acpx_demo_gemini_uplift.py
-- 0074_simplify_demo_prompts.py
-- 0075_add_acpx_tool_surface.py
-- 0076_add_acpxer.py
-- 0077_add_whatstlatamini.py
-- 0078_add_chat_agent_keyboard_tool.py
-- 0079_add_chat_agent_mouse_tool.py
-- 0080_add_chat_agent_sleeper_tool.py
-- 0081_add_window_present_and_run_wait_tools.py
```

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```
-- 0082_add_chat_agent_j_decompiler_tool.py
-- 0083_add_de_compressor.py
-- 0084_add_chat_agent_de_compressor_tool.py
-- 0085_add_unrealer.py
-- 0086_add_chat_agent_unrealer_tool.py
-- 0087_add_unrealer_demo_prompt.py
-- 0088_add_reviewer.py
-- 0089_add_analyzer.py
-- 0090_add_reviewer_analyzer_demo_prompts.py
-- 0091_add_playwrighter.py
-- 0092_add_chat_agent_playwrighter_tool.py
-- 0093_add_windower.py
-- 0094_add_chat_agent_windower_tool.py
-- 0095_add_windower_playwrighter_demo_prompts.py
-- 0096_add_director_virtuoso_demo_prompts.py
-- 0097_add_kalier.py
-- 0098_add_chat_agent_kalier_tool.py
-- 0099_add_kalier_demo_prompts.py
-- 0100_add_unrealer_extended_demo_prompts.py
-- 0101_add_stm32er.py
-- 0102_add_chat_agent_stm32er_tool.py
-- 0103_add_stm32er_demo_prompts.py
-- __init__.py
-- monitors/
--   |-- monitor_sql.py
-- opus_client/
--   |-- claude_opus_client.py
--   |-- examples.py
--   |-- README.md
-- rag/
--   |-- __init__.py
--   |-- config.py
--   |-- factory.py
--   |-- interaction.py
--   |-- interface.py
--   |-- loaders.py
--   |-- prompts.py
--   |-- retrieval.py
--   |-- splitters.py
--   |-- utils.py
--   |-- chains/
--   |   |-- base.py
--   |   |-- basic.py
--   |   |-- history_aware.py
--   |   |-- unified.py
-- services/
--   |-- __init__.py
--   |-- agent_contracts.py
--   |-- agent_paths.py
--   |-- answer_analyzer.py
--   |-- filesystem.py
--   |-- flow_compiler.py
--   |-- flow_spec.py
--   |-- response_parser.py
-- skills/
--   |-- __init__.py
--   |-- frontmatter.py
--   |-- harness.py
--   |-- io_contract.py
--   |-- registry.py
-- skills_pkg/
--   |-- __meta/
--   |   |-- lint.py
--   |   |-- schema.json
--   |-- acp_router/
--   |   |-- SKILL.md
--   |-- code_review/
--   |   |-- SKILL.md
--   |-- github/
--   |   |-- SKILL.md
--   |-- gmail/
--   |   |-- SKILL.md
--   |-- hello_world/
--   |   |-- SKILL.md
--   |-- jira/
--   |   |-- SKILL.md
```

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```
-- kali_pentest/
--   |-- SKILL.md
--   |-- notion/
--   |   |-- SKILL.md
--   |-- security_audit/
--   |   |-- SKILL.md
--   |-- setup_new_acpx_key/
--   |   |-- SKILL.md
--   |-- skill_creator/
--   |   |-- SKILL.md
--   |   |-- scripts/
--   |       |-- quick_validate.py
--   |-- slack/
--   |   |-- SKILL.md
--   |-- summarize/
--   |   |-- SKILL.md
--   |-- tlamadini_allowed_hosts_tighten/
--   |   |-- SKILL.md
--   |-- tlamadini_csrf_exempt_audit/
--   |   |-- SKILL.md
--   |-- tlamadini_exec_report_row_adder/
--   |   |-- SKILL.md
--   |-- tlamadini_flow_from_objective/
--   |   |-- SKILL.md
--   |-- tlamadini_flw_doctor/
--   |   |-- SKILL.md
--   |-- tlamadini_new_acp_agent/
--   |   |-- SKILL.md
--   |-- tlamadini_planner_trace_replay/
--   |   |-- SKILL.md
--   |-- tlamadini_static_version_bumper/
--   |   |-- SKILL.md
--   |-- todoist/
--   |   |-- SKILL.md
--   |-- trello/
--   |   |-- SKILL.md
--   |-- weather/
--   |   |-- SKILL.md
-- static/
--   |-- agent/
--   |   |-- css/
--   |   |   |-- agent_page.css
--   |   |   |-- agentic_control_panel.css
--   |   |   |-- login.css
--   |   |   |-- skills_dialog.css
--   |   |   |-- tools_dialog.css
--   |   |   |-- welcome.css
--   |   |-- img/
--   |   |   |-- spinner.svg
--   |   |   |-- Tlamadini.ico
--   |   |-- js/
--   |       |-- acp-agent-connectors.js
--   |       |-- acp-canvas-core.js
--   |       |-- acp-canvas-undo.js
--   |       |-- acp-control-buttons.js
--   |       |-- acp-file-io.js
--   |       |-- acp-flow-snapshot.js
--   |       |-- acp-globals.js
--   |       |-- acp-layout.js
--   |       |-- acp-parametrizer-dialog.js
--   |       |-- acp-running-state.js
--   |       |-- acp-session.js
--   |       |-- acp-undo-manager.js
--   |       |-- acp-validate.js
--   |       |-- agent_page_canvas.js
--   |       |-- agent_page_chat.js
--   |       |-- agent_page_context.js
--   |       |-- agent_page_dialogs.js
--   |       |-- agent_page_init.js
--   |       |-- agent_page_layout.js
--   |       |-- agent_page_state.js
--   |       |-- agent_page_ui.js
--   |       |-- agentic_control_panel.js
--   |       |-- canvas_item_dialog.js
--   |       |-- chat_page_runtime_poller.js
--   |       |-- contextual_menus.js
```


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```
| | | |-- shared-runtime-dialogs.js
| | | |-- skills_dialog.js
| | | |-- tools_dialog.js
| | | |-- sounds/
| | | | |-- hypervisor_alert.wav
| | | | |-- notification.wav
| | | |-- video/
| | | | |-- XAIHT-Tlamatini.mp4
| | |-- telegram/
| | | |-- AztecTlamatini.txt
| | | |-- config.yaml
| | | |-- telegram_receiver.py
| | |-- templates/
| | | |-- agent/
| | | | |-- agent_page.html
| | | | |-- agentic_control_panel.html
| | | | |-- login.html
| | | | |-- welcome.html
| | |-- TlamatiniSourceCode/
| | | |-- README.md
|-- DB/
| |-- Older/
| | |-- README.md
| |-- ToLoad/
| | |-- README.md
|-- jd-cli/
| |-- jd-cli.bat
| |-- jd-cli.jar
|-- tlamatini/
| |-- __init__.py
| |-- asgi.py
| |-- context_processors.py
| |-- logging_filters.py
| |-- middleware.py
| |-- settings.py
| |-- urls.py
| |-- wsgi.py
```